

even use Newton's laws of physics to simulate the flight parameters of an actual spaceship. That might be an overstatement but you know what we're talking about.



Bouncing bubbles simulated using Processing

Processing was developed by Casey Reas and Benjamin Fry from the Aesthetics and Computation Group at the MIT Media Lab in 2001. Their main goal was to create a tool to get beginners started with programming through the instant gratification of getting a visual feedback. The

language is built on top of Java but makes use of a simplified syntax. The same goes for the graphics programming model which depends on many libraries that help achieve the graphical interface.

Significance

It is a programming language that uses a simplified Java syntax taking away the hassle of understanding the actual Java syntax which for beginners, is quite complex. And it still maintains the functionality necessary to make visualizations and perform other necessary operations. This makes it a great tool to start programming, to create graphics and animations with visual analysis of data which in turn helps create physics simulations.

The core Processing framework simplifies the use of basic multimedia libraries (OpenGL, PDF, camera capture), thus, removing the overhead involved in the setting up of basic applications. It uses an extensible code structure to allow the creation of dozens of useful libraries for everything from 3D import / export tools to ones that create complex 3D geometrical figures.

For scientific data analysis, it can act as tool that combines data from multiple sources and helps visualise the data. Basically, it makes for easy comparison of data from various sensors and other hardware.

Processing the Sketch (Coding)

To work with graphics in Processing, we use the Coordinate system which is used for positioning every shape. The computer's coordinate system is different from that of the Cartesian coordinate system taught in schools.

The difference isn't that big and the image on the left will help you understand the difference between the Cartesian and computer coordinate system.